

CO2-Assesement Tool-01

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Course Code: Csa1214

Course: Computer architecture

1. 2's Complement (+45 and -23)

Step 1: Convert to binary

+45 = 00101101

+23 = 00010111

Step 2: Find 2's complement of 23

1's complement = 11101000

+1 = 11101001 (-23)

Step 3: Addition (+45 + (-23))

00101101

+ 11101001

00010110

Result = 22

Step 4: Subtraction (+45 - (-23))

00101101

+ 00010111

01000100

Result = 68

Overflow: No.

2. Carry Look-Ahead Adder (CLA)

Example:

A = 1010

B = 0111

Step 1: Find Generate ($G = A \cdot B$)

Bit G

0 0

1 1

2 0

3 0

Step 2: Find Propagate ($P = A \oplus B$)

Bit P

0 1

1 0

2 1

3 1

Step 3: Compute carries

- $C1 = G0 + P0C0$
- $C2 = G1 + P1G0 + P1P0C0$
- $C3 = G2 + \dots$

Step 4: Find Sum

CLA computes all carries together, making it faster than Ripple Carry Adder.

3. Booth's Algorithm ($-6 \times +5$)

Step 1: Binary values

$M = 1010$ (-6)

$Q = 0101$ ($+5$)

$A = 0000$

$Q-1 = 0$

Step 2: Check $Q0$ and $Q-1$

- $10 \rightarrow A = A - M$
- $01 \rightarrow A = A + M$
- 00 or $11 \rightarrow$ No operation

Step 3: Perform Arithmetic Right Shift

Repeat for 4 cycles.

Step 4: Final Result

AQ = 11100010

= -30

4. Division ($13 \div 3$)

13 = 1101

3 = 0011

Restoring Division

Step 1: Shift left.

Step 2: Subtract divisor.

Step 3: If remainder is negative, restore by adding divisor.

Step 4: Set quotient bit.

Final Answer

Quotient = 0100 (4)

Remainder = 0001 (1)

Non-Restoring Division

Step 1: Shift left.

Step 2: Add/Subtract divisor based on previous remainder.

Step 3: Repeat for all bits.

Step 4: Restore only at the end if required.

Result

Quotient = 4

Remainder = 1

5. IEEE 754 (-13.25)

Step 1: Convert to binary

13.25 = 1101.01

Step 2: Normalize

$$1.10101 \times 2^3$$

Step 3: Find fields

- Sign = 1
- Exponent = $3 + 127 = 130 = 10000010$
- Mantissa = 101010000000000000000000

Step 4: Single Precision

1 | 10000010 | 101010000000000000000000

Step 5: Double Precision

- Sign = 1
- Exponent = 10000000010
- Mantissa = 101010...

Floating-Point Addition Steps

1. Align exponents.
2. Add mantissas.
3. Normalize.
4. Round.
5. Store final result.